A Colored Turán Proof of the Lollipop Formula

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June 21, 2026

Abstract

A lollipop is a circle together with the outward radial ray beginning at one point of the circle. Let $a_{\mathcal{L}}(n)$ be the maximum number of connected components of the complement of a union of n lollipops in the plane. We prove, for every $n \geq 0$,

$$a_{\mathcal{L}}(n) = 4\binom{n}{2} + S(n) + n + 1,$$

where

$$S(n) = \max_{\substack{a+b+c+d=n \\ 0\leq a\leq b\leq c\leq d}} \left(3 \sum_{1\leq i<j\leq 4} m_i m_j - 2ab\right), \qquad (m_1, m_2, m_3, m_4) = (a, b, c, d).$$

The upper bound is valid without a general-position assumption. Its geometric input is encoded by two graphs: non-close pairs form a K_4-free graph and non-intriguing pairs form a K_5-free graph. A colored Zykov symmetrization reduces the resulting extremal problem to a weighted blocker inequality. The final inequality is an exact 3×4 integer-matrix problem, proved by support compression to star forests. For the lower bound, we give a polynomial one-parameter family in which every two distinct nearby copies have exactly four transverse intersections, and blow up an all-rational perturbation of Karlsson's four-lollipop configuration. Every decisive inequality for the base configuration is checked by exact rational arithmetic; no floating-point branch or sign decision is used.

1 Statement and notation

For $c \in \mathbb{R}^2$, $r > 0$, and a unit vector $u \in S^1$, write

$$C(c, r) = \{x \in \mathbb{R}^2 : |x - c| = r\}, \qquad R(c, r, u) = \{c + tu : t \geq r\}.$$

The associated *lollipop* is

$$L(c, r, u) = C(c, r) \cup R(c, r, u),$$

and its *anchor* is $a = c + ru$. No general-position hypothesis is part of the definition of $a_{\mathcal{L}}(n)$.

For $n \geq 0$, define

$$S(n) = \max_{\substack{a+b+c+d=n \\ 0\leq a\leq b\leq c\leq d}} \left(3 \sum_{1\leq i<j\leq 4} m_i m_j - 2ab\right), \qquad (m_1, m_2, m_3, m_4) = (a, b, c, d). \tag{1}$$

Equivalently, for a fixed multiset of four cluster sizes, the penalized product is the product of the two smallest sizes.

Theorem 1.1. *For every integer $n \geq 0$,*

$$a_{\mathcal{L}}(n) = 4\binom{n}{2} + S(n) + n + 1. \tag{2}$$

The proof separates into topology, pair geometry, and an exact colored extremal theorem. Cutler-Karlsson-Sloane established the first geometric bounds and the exact cases through four lollipops; Paulsen introduced the close-pair and circle-angle obstructions; and Mahajan-Chopra recast those obstructions as a two-graph problem, determining all cases through 17 and also $n = 19$ while leaving one-region gaps at 18 and 20 [1, 2, 4]. Theorem 1.1 closes those gaps and proves the resulting formula for every n.

The topological argument below is included in order to remove a frequently used but unnecessary assertion that every extremizer may simply be assumed generic.

2 A region bound valid for arbitrary arrangements

Let $S^2 = \mathbb{R}^2 \cup \{\infty\}$ be the one-point compactification. For a lollipop L, put $\widehat{L} = L \cup \{\infty\}$. The compactified stem is a closed arc from the anchor to ∞, so $\widehat{L}$ is connected and has first Betti number one. All homology in this section has coefficients in $\mathbb{Q}$.

For two lollipops define

$$q(L, M) = \beta_0(\widehat{L} \cap \widehat{M}) - 1. \tag{3}$$

The subtraction removes the component containing ∞, which is present for every pair. Thus $q(L, M)$ counts the other connected components of the compactified pair intersection, not intersection points with multiplicity. This definition remains meaningful for tangencies, coincident circles, and overlapping stems.

Proposition 2.1 (topological region inequality)**.** *For arbitrary lollipops $L_1, \dots, L_n$, the number F of connected components of $\mathbb{R}^2 \setminus \bigcup_i L_i$ satisfies*

$$F \leq n + 1 + \sum_{1 \leq i < j \leq n} q(L_i, L_j). \tag{4}$$

If the arrangement has only transverse pairwise crossings, no triple point, no crossing at an anchor, and no parallel or overlapping stems, then equality holds and $q(L_i, L_j)$ is the ordinary number of pairwise crossings.

Proof. Realize S^2 as the unit sphere in $\mathbb{R}^3$. Under inverse stereographic projection, circles, rays, and the point at infinity are semialgebraic; hence the resulting finite family admits a compatible semialgebraic triangulation. In particular, every union and intersection below is a compact subcomplex; its Betti numbers and the number of complement components are finite [5]. Put $X_k = \bigcup_{i=1}^{k} \widehat{L}_i$. Since every $\widehat{L}_i$ contains ∞, each X_k is connected. The Mayer–Vietoris exact sequence for $X_{k-1} \cup \widehat{L}_k$ gives

$$\beta_1(X_k) \leq \beta_1(X_{k-1}) + \beta_1(\widehat{L}_k) + \beta_0(X_{k-1} \cap \widehat{L}_k) - 1.$$

As $\beta_1(\widehat{L}_k) = 1$, this becomes

$$\beta_1(X_k) \leq \beta_1(X_{k-1}) + \beta_0(X_{k-1} \cap \widehat{L}_k). \tag{5}$$

Now

$$X_{k-1} \cap \widehat{L}_k = \bigcup_{j<k} (\widehat{L}_j \cap \widehat{L}_k).$$

All sets in this union contain ∞. A union of sets with one common component has at most one common component plus the sum of their remaining component counts. Consequently

$$\beta_0(X_{k-1} \cap \widehat{L}_k) \le 1 + \sum_{j<k} q(L_j, L_k).$$

Starting from $\beta_1(X_1) = 1$ and iterating (5) yields

$$\beta_1(X_n) \le n + \sum_{i<j} q(L_i, L_j). \tag{6}$$

Alexander duality on S^2 [6, Section 3.3] gives

$$F - 1 = \dim \widetilde{H}_0(S^2 \setminus X_n; \mathbb{Q}) = \dim \widetilde{H}^1(X_n; \mathbb{Q}) = \beta_1(X_n),$$

which proves (4). The case $n = 0$ is immediate.

For a generic arrangement, X_n is a finite connected graph. If C is the total number of finite crossings, its vertices are the C crossings, the n anchors, and ∞, so $V = C + n + 1$. Counting circle and stem segments gives $E = 2C + 2n$: each lollipop contributes one baseline circle edge and one baseline stem edge, and every crossing contributes one subdivision on each of two lollipops. Hence $\beta_1 = E - V + 1 = C + n$, and therefore $F = C + n + 1$. In this case every finite pair-intersection component is one crossing, so equality also follows in (4). □

3 Pair geometry and the two forced obstructions

For two lollipops $L_i = L(c_i, r_i, u_i)$, set $d = c_2 - c_1$. They are called *close* if

$$u_1 \cdot u_2 \ge 0, \tag{7}$$

that is, the smaller angle between their stem directions is at most $\pi/2$. They are called *intriguing* if either their two circle curves are disjoint, or

$$|d|^2 \le r_1^2 + r_2^2. \tag{8}$$

When the circles meet transversely, (8) says that the angle between equally oriented circle tangents is at most $\pi/2$.

We first establish all pair bounds, including degenerate cases, directly.

Lemma 3.1 (mixed intersections for a close pair)**.** *If $u_1 \cdot u_2 \ge 0$, then*

$$\#(R(c_1, r_1, u_1) \cap C(c_2, r_2)) + \#(R(c_2, r_2, u_2) \cap C(c_1, r_1)) \le 2.$$

Proof. A ray meets a circle in at most two points. Suppose the first mixed intersection has two distinct points. Along the supporting line of the first ray the two intersection parameters have average $d \cdot u_1$. Both parameters are at least r_1, and they are distinct, so $d \cdot u_1 > r_1$.

If $c_2 + tu_2 \in C(c_1, r_1)$ for some $t \ge r_2 > 0$, then

$$d \cdot u_1 + t(u_2 \cdot u_1) = (c_2 + tu_2 - c_1) \cdot u_1 \le |c_2 + tu_2 - c_1| = r_1.$$

Under (7) this contradicts $d \cdot u_1 > r_1$. Thus two intersections in one mixed component force zero in the reverse component. If neither mixed component has two points, each has at most one, proving the claim. □

Lemma 3.2 (mixed intersections for intriguing circles). *Assume the circles meet and* $|d|^2 \le r_1^2 + r_2^2$.

(i) Each set $R(c_i, r_i, u_i) \cap C(c_j, r_j)$ *contains at most one point outside* $C(c_1, r_1) \cap C(c_2, r_2)$.

(ii) If both mixed sets contain such a point, then the two stems have no finite intersection.

Proof. Let $p = d \cdot u_1$ and write $|d|^2 = p^2 + h^2$, where h is the perpendicular distance from c_2 to the first supporting line. If the first ray meets the second circle in two distinct points with parameters $t_- < t_+$, then

$$t_\pm = p \pm \sqrt{r_2^2 - h^2}, \qquad t_- \ge r_1.$$

Put $z = p - r_1$. We have $z > 0$ and $z \ge \sqrt{r_2^2 - h^2}$. Therefore

$$|d|^2 - r_1^2 - r_2^2 = 2r_1 z + z^2 - (r_2^2 - h^2) > 0,$$

contrary to the hypothesis. Thus there are not two mixed points.

A singleton mixed intersection outside the common circle intersection cannot be a tangency. Indeed, at a tangent parameter $t = p \ge r_1$ one has $|d|^2 = p^2 + r_2^2$; the intriguing inequality forces $p \le r_1$, hence $p = r_1$, and the tangent point is the first anchor and lies on both circles. Consequently a singleton outside the common circle intersection is a simple root beyond the anchor. Since the quadratic distance function tends to $+\infty$ and has exactly one root after the anchor, its value at the anchor is negative. In other words, the first anchor lies strictly inside the second disk. This proves (i), and gives the corresponding interior statement for either orientation.

Assume now that both outside mixed points exist. Let $a_i = c_i + r_i u_i$ be the anchors. We have $|a_1 - c_2| < r_2$ and $|a_2 - c_1| < r_1$. If the two rays met at a finite point, write

$$x = c_1 + (r_1 + s)u_1 = c_2 + (r_2 + t)u_2, \qquad s, t \ge 0.$$

Because a_1 is inside the second disk while every point of the second stem is outside that disk, $s > 0$; similarly $t > 0$. Substituting for $c_2 - c_1$ and using $u_1 \cdot u_2 \le 1$ gives

$$|a_1 - c_2|^2 = |-su_1 + (r_2 + t)u_2|^2 \ge (r_2 + t - s)^2.$$

The strict inequality $|a_1 - c_2| < r_2$ therefore implies $t < s$. Interchanging 1 and 2 gives $s < t$, a contradiction. This proves (ii). ☐

Proposition 3.3 (pair savings). *For arbitrary lollipops* L, M,

$$\begin{aligned} q(L, M) &\le 7, \\ L, M \text{ close} \Longrightarrow q(L, M) &\le 5, \\ L, M \text{ intriguing} \Longrightarrow q(L, M) &\le 5, \\ L, M \text{ close and intriguing} \Longrightarrow q(L, M) &\le 4. \end{aligned}$$

Proof. The circle–circle intersection has at most two connected components (two points, one point, one coincident circle, or none). Each mixed ray–circle intersection has at most two points. The finite part of a ray–ray intersection contributes at most one component not connected to ∞. This gives $q \le 2 + 2 + 2 + 1 = 7$.

For a close pair, Lemma 3.1 reduces the total mixed contribution to at most two, giving $q \le 2+2+1 = 5$.

If an intriguing pair has disjoint circles, the circle–circle contribution vanishes and the general count gives $q \le 0 + 2 + 2 + 1 = 5$. Otherwise the circles meet and satisfy (8). Let $m_{12}, m_{21} \in \{0, 1\}$ count mixed points outside the common circle intersection. Mixed points lying on both circles do not create additional components beyond the circle–circle components. By Lemma 3.2, if $m_{12} = m_{21} = 1$ then the finite ray–ray contribution is zero; otherwise it is at most one. Hence in either case

$$q \le 2 + m_{12} + m_{21} + \mathbf{1}_{\{\text{finite ray–ray component}\}} \le 4.$$

Thus an intersecting intriguing pair actually satisfies the stronger bound $q \le 4$. Finally, a pair that is both close and intriguing has bound at most four in the intersecting case, and at most $0+2+1 = 3$ in the disjoint case. □

The two Ramsey-type consequences are next.

Lemma 3.4 (forced close pair)**.** *Among any four stem directions there is a close pair.*

Proof. Place the four directions in cyclic order on S^1. Their four cyclic gaps sum to 2π, so one gap is at most $\pi/2$. □

Lemma 3.5 (forced intriguing pair)**.** *Among any five circles there is an intriguing pair.*

Proof. Suppose not. Then every two circle curves meet, and for their center distance $d_{ij} = |c_i - c_j|$ one has

$$r_i^2 + r_j^2 < d_{ij}^2 \le (r_i + r_j)^2. \tag{9}$$

External tangencies can be removed uniformly: because the family is finite, there is a number $\lambda > 1$, arbitrarily close to one, such that replacing all r_i by λr_i preserves $d_{ij}^2 > \lambda^2(r_i^2 + r_j^2)$ and $d_{ij} > \lambda|r_i - r_j|$, while making $d_{ij} < \lambda(r_i + r_j)$. Thus the enlarged circles meet transversely and at an obtuse equally oriented angle. We rename their radii r_i.

For each circle set

$$v_i = \frac{1}{r_i}(1, r_i^2 - |c_i|^2, c_i) \in \mathbb{R}^4$$

and equip triples (α, β, x) with the symmetric bilinear form

$$\langle(\alpha, \beta, x), (\alpha', \beta', x')\rangle = \frac{1}{2}(\alpha\beta' + \beta\alpha') + x \cdot x'.$$

A direct calculation gives

$$\langle v_i, v_i\rangle = 1, \qquad \langle v_i, v_j\rangle = \frac{r_i^2 + r_j^2 - d_{ij}^2}{2r_ir_j} \in (-1, 0).$$

The five vectors are linearly dependent. In a nonzero relation, positive and negative coefficients both occur because every first coordinate is positive. After changing the sign if necessary, write

$$\sum_{i\in P} a_i v_i = \sum_{j\in N} b_j v_j, \qquad a_i, b_j > 0,$$

with $1 \le |P| \le 2$. Put $w = \sum_{i\in P} v_i$. If $P = \{i\}$, the inner product of the left side with w is $a_i > 0$. If $P = \{i, k\}$, it is $(a_i + a_k)(1 + \langle v_i, v_k\rangle) > 0$. The inner product of the right side with w is strictly negative, because every cross inner product is negative. This contradiction proves the lemma. □

4 The two-graph reduction

Given an arbitrary arrangement, define graphs D, E on the lollipops by

$$ij \in D \iff L_i, L_j \text{ are not close}, \qquad ij \in E \iff L_i, L_j \text{ are not intriguing}.$$

Lemmas 3.4 and 3.5 imply

$$D \text{ is } K_4\text{-free}, \qquad E \text{ is } K_5\text{-free}. \tag{10}$$

Proposition 3.3 gives, for every pair,

$$q(L_i, L_j) \le 4 + \mathbf{1}_{ij\in D} + \mathbf{1}_{ij\in E} + \mathbf{1}_{ij\in D\cap E}. \tag{11}$$

Indeed, the four cases have respective bounds $4, 5, 5, 7$. Put

$$\sigma(D, E) = |D| + |E| + |D \cap E|.$$

Combining Proposition 2.1 with (11),

$$F \le 4\binom{n}{2} + \sigma(D, E) + n + 1. \tag{12}$$

Thus the upper bound in Theorem 1.1 follows from the next purely combinatorial theorem.

Theorem 4.1 (colored Turán theorem)**.** *If D is K_4-free and E is K_5-free on the same n-vertex set, then*

$$\sigma(D, E) \le S(n). \tag{13}$$

5 Colored Zykov symmetrization

Color each unordered pair by

$$0: \text{ neither graph}, \quad A : E \setminus D, \quad B : D \setminus E, \quad X : D \cap E.$$

The weights are $0, 1, 1, 3$, respectively, and their total is $\sigma(D, E)$.

Lemma 5.1 (colored Zykov reduction)**.** *An extremal feasible coloring can be chosen as a blow-up of a complete nonzero-colored quotient: the vertices are partitioned into classes $V_1, \dots, V_k$, pairs inside a class have color 0, and every pair between two classes has one fixed color in $\{A, B, X\}$.*

Proof. Call two vertices *zero-twins* if they are equal, or their mutual color is 0 and their colored neighborhoods agree outside the pair. This is an equivalence relation. Among all feasible colorings of maximum weight, choose one maximizing the sum of squares of the zero-twin class sizes.

Suppose distinct zero-twin classes P, Q are joined by color 0. Then every P–Q pair has color 0. Let $d_w(P)$ be the weighted degree of a vertex of P to vertices outside $P \cup Q$, and define $d_w(Q)$ similarly. If $d_w(P) \le d_w(Q)$, replace every vertex of P by a clone of a representative of Q, keeping all pairs inside $P \cup Q$ color 0. The total weight changes by

$$|P|(d_w(Q) - d_w(P)) \ge 0.$$

No new K_4 in D or K_5 in E is created: a new clique containing a clone from P but no vertex of Q is obtained from an old clique by replacing the clone by its model in Q, while a clique cannot use vertices from both P and Q because their mutual pairs have color 0. If the weighted degrees are unequal, extremality is contradicted; if they are equal, P and Q merge and the square-sum tie-breaker is contradicted. The opposite inequality is symmetric. Hence no zero-colored pair joins distinct classes, and the zero-twin definition makes every cross-class color constant. $\square$

Let the quotient class sizes be $x_1, \dots, x_k$, with $\sum_i x_i = n$, and put

$$Q = \sum_i x_i^2, \qquad P = \sum_{i<j} x_i x_j = \frac{n^2 - Q}{2}.$$

Let a and b be the total weighted masses $\sum x_i x_j$ on quotient edges of colors A and B. Since all other cross-edges have color X,

$$\sigma = 3P - 2a - 2b = \frac{3}{2}(n^2 - Q) - 2a - 2b. \tag{14}$$

Because D consists of colors B, X, every four quotient vertices contain an A-edge; hence $\alpha(A) \le 3$. Similarly $\alpha(B) \le 4$.

6 Weighted blockers

For nonnegative integer weights $x_1, \dots, x_k$ of total n, and a graph G on $[k]$, write

$$w(G) = \sum_{ij \in E(G)} x_i x_j.$$

For $r \ge 1$, define

$$\rho_r(x) = \min_{\mathcal{P}} \sum_{P \in \mathcal{P}} \left(\sum_{i \in P} x_i \right)^2,$$

where $\mathcal{P}$ ranges over partitions into r possibly empty parts.

Lemma 6.1 (weighted Turán theorem)**.** *If G is K_{r+1}-free, then*

$$w(G) \le \frac{1}{2}(n^2 - \rho_r(x)).$$

Proof. Zero-weight vertices may be deleted, so assume every $x_i > 0$. Call two nonadjacent vertices twins when their open neighborhoods agree, and partition the vertices into twin classes. Among all maximum-weight K_{r+1}-free graphs, choose one maximizing the sum of the squares of the total weights of its twin classes.

Suppose two distinct twin classes P, Q are nonadjacent. Let $d_w(P)$ and $d_w(Q)$ denote the weighted degrees of their vertices outside $P \cup Q$; these are constant on each class. If $d_w(P) \le d_w(Q)$, replace every vertex of P by a clone of a representative of Q, keeping all pairs in $P \cup Q$ nonadjacent. The edge weight changes by

$$\left(\sum_{u \in P} x_u \right) (d_w(Q) - d_w(P)) \ge 0.$$

The operation preserves K_{r+1}-freeness: a new clique cannot use vertices from both P and Q, and a clique using a clone from P maps to an old clique by replacing that clone with its model in Q. Strict inequality contradicts maximality; equality merges P and Q and strictly increases the tie-breaker. The opposite ordering is symmetric. Hence distinct twin classes are pairwise adjacent.

The graph is therefore complete multipartite, and K_{r+1}-freeness leaves at most r parts. If their total weights are $s_1, \dots, s_r$, its edge weight is

$$\sum_{i<j} s_i s_j = \frac{1}{2}\left(n^2 - \sum_i s_i^2\right).$$

Minimizing the final square-sum over partitions of the original vertex weights is exactly the definition of $\rho_r(x)$. □

Define

$$M(n) = \min_{\substack{a+b+c+d=n \\ a,b,c,d\in\mathbb{Z}_{\geq 0}}} \left[\frac{3}{2}(a^2+b^2+c^2+d^2) + 2ab\right]. \tag{15}$$

For a fixed multiset, the $2ab$ term is minimized by assigning a, b to the two smallest entries. Hence

$$S(n) = \frac{3}{2}n^2 - M(n). \tag{16}$$

Lemma 6.2 (weighted blocker lemma)**.** *If $\alpha(A) \leq 3$ and $\alpha(B) \leq 4$, then*

$$\frac{3}{2}Q + 2w(A) + 2w(B) \geq M(n). \tag{17}$$

Proof. The complement of A is K_4-free. Lemma 6.1 gives

$$\frac{1}{2}(n^2 - Q) - w(A) \leq \frac{1}{2}(n^2 - \rho_3(x)), \qquad w(A) \geq \frac{1}{2}(\rho_3(x) - Q).$$

Likewise $w(B) \geq \frac{1}{2}(\rho_4(x) - Q)$. Therefore

$$\frac{3}{2}Q + 2w(A) + 2w(B) \geq \rho_3(x) + \rho_4(x) - \frac{1}{2}Q. \tag{18}$$

Choose partitions attaining ρ_3 and ρ_4, and intersect their parts. Let $U = (u_{ij})$ be the resulting 3×4 nonnegative integer matrix, with row sums r_i and column sums c_j. Then

$$\rho_3 = \sum_i r_i^2, \qquad \rho_4 = \sum_j c_j^2, \qquad Q \leq \sum_{i,j} u_{ij}^2,$$

the last inequality following from $\sum z_\ell^2 \leq (\sum z_\ell)^2$ in each cell. Thus the right side of (18) is at least

$$F(U) := \sum_i r_i^2 + \sum_j c_j^2 - \frac{1}{2}\sum_{i,j} u_{ij}^2.$$

The matrix theorem in Section 7 proves $F(U) \geq M(n)$. □

Proof of Theorem 4.1. Use Lemma 5.1 and the quotient notation above. The blocker lemma implies $\frac{3}{2}Q + 2a + 2b \geq M(n)$. Equation (14) and (16) give

$$\sigma = \frac{3}{2}n^2 - \left(\frac{3}{2}Q + 2a + 2b\right) \leq \frac{3}{2}n^2 - M(n) = S(n).$$

□

7 The exact 3×4 matrix inequality

Theorem 7.1 (matrix theorem)**.** *For every 3×4 nonnegative integer matrix U of total sum n,*

$$F(U) = \sum_i r_i^2 + \sum_j c_j^2 - \frac{1}{2} \sum_{i,j} u_{ij}^2 \geq M(n). \tag{19}$$

View positive cells as weighted edges of $K_{3,4}$. Expanding row and column squares gives

$$F(U) = \frac{3}{2} \sum_e u_e^2 + 2 \sum_{e \sim f} u_e u_f, \tag{20}$$

where adjacency means sharing a row or column vertex.

Lemma 7.2 (support compression)**.** *Every nonnegative integer matrix U can be replaced by a matrix U' of the same total sum, with $F(U') \leq F(U)$, whose support graph is a star forest.*

Proof. If the support contains a cycle, add ε and $-\varepsilon$ alternately around it. All row and column sums remain fixed, while $-\frac{1}{2}\sum u_{ij}^2$ is a concave quadratic in ε. The allowed integer values form a finite interval containing zero. At least one endpoint has value no larger than at zero and makes an entry vanish. Repetition removes all cycles.

Suppose next that a tree component has diameter at least four. On four consecutive path edges of masses x_1, x_2, x_3, x_4, use

$$(x_1, x_2, x_3, x_4) \mapsto (x_1 + \varepsilon, x_2 - \varepsilon, x_3 + \varepsilon, x_4 - \varepsilon).$$

The three internal vertex sums are fixed and the two endpoint-square terms contribute $2\varepsilon^2$. The four cell-square terms contribute $-\frac{1}{2} \cdot 4\varepsilon^2 = -2\varepsilon^2$. Hence F is affine in ε on the allowed integer interval. An endpoint does not increase F and deletes an edge. Repetition leaves a forest of diameter at most three.

A diameter-three component is a double star. Let its central edge have mass y. Choose leaf masses x, z on opposite sides, and let A, B be the sums of the other leaf masses at the respective centers. Deleting the central edge and adding y to the x-edge gives, by (20),

$$F_{\text{old}} - F_x = y(2B + 2z - x).$$

Adding y to the z-edge instead gives

$$F_{\text{old}} - F_z = y(2A + 2x - z).$$

Both expressions cannot be negative, since that would imply $x > 2B + 2z \geq 2z$ and $z > 2A + 2x \geq 2x$. One move therefore does not increase F and deletes the central edge. Repetition removes every diameter-three component. A forest of diameter at most two is a star forest. □

For a star of degree t, edge masses $z_1, \ldots, z_t$, and total mass $s = \sum z_i$, its contribution is

$$s^2 + \frac{1}{2} \sum_i z_i^2 \geq \left(1 + \frac{1}{2t}\right) s^2 = \frac{s^2}{\eta_t}, \qquad \eta_t = \frac{2t}{2t+1}. \tag{21}$$

Lemma 7.3 (star-forest bound)**.** *Every star-forest matrix U of total mass n satisfies $F(U) \geq M(n)$.*

Proof. If $n = 0$, then $U = 0$ and $F(U) = M(0) = 0$. Assume henceforth that $n > 0$. Let the nontrivial star components have degrees t_ν and masses s_ν. Except for degree list $(2,1,1)$,

$$\Gamma := \sum_\nu \eta_{t_\nu} \le 2. \tag{22}$$

Indeed, there are at most three nontrivial components because each uses a row vertex. One or two components give $\Gamma < 2$. With three components, their vertex counts satisfy $\sum(t_\nu + 1) \le 7$, so the only degree lists are $(1,1,1)$ and $(2,1,1)$; the former has $\Gamma = 2$.

For all nonexceptional degree lists, (21) and Cauchy's inequality give

$$F(U) \ge \sum_\nu \frac{s_\nu^2}{\eta_{t_\nu}} \ge \frac{n^2}{\Gamma} \ge \frac{n^2}{2}.$$

For $n \ge 4$, $M(n) \le n^2/2$: if $n = 4q + r$, use respectively

$$(q,q,q,q),\ (q,q,q,q+1),\ (q,q,q+1,q+1),\ (q,q+1,q+1,q+1)$$

for $r = 0,1,2,3$ in (15); direct substitution gives the claimed inequality. For $1 \le n \le 3$, the inequality $z^2 \ge z$ for $z \in \mathbb{Z}_{\ge 0}$ gives $M(n) \ge 3n/2$, and the quadruples displayed in Section 9 attain equality. Meanwhile

$$F(U) = \frac{3}{2}n + 3\sum_{i,j}\binom{u_{ij}}{2} + 2\sum_{e\sim f} u_e u_f \ge \frac{3}{2}n.$$

It remains to treat degree list $(2,1,1)$. The degree-two star must be centered at a row: if centered at a column it uses two rows, leaving only one row for two disjoint edges. After row and column permutations the positive entries are

$$\begin{pmatrix} a & b & 0 & 0 \\ 0 & 0 & c & 0 \\ 0 & 0 & 0 & d \end{pmatrix}.$$

Then

$$F(U) = \frac{3}{2}(a^2+b^2+c^2+d^2) + 2ab \ge M(n)$$

by the definition of $M(n)$. □

Proof of Theorem 7.1. Compress U to a star-forest matrix U' by Lemma 7.2. Then $F(U) \ge F(U') \ge M(n)$ by Lemma 7.3. □

Combining Theorem 4.1 with (12) proves the upper bound in Theorem 1.1 for all arrangements, including degenerate ones.

8 The lower construction

We now construct generic arrangements attaining the upper bound. Two stability lemmas are stated explicitly because stems are unbounded.

8.1 An all-rational four-lollipop base

Write $Q_{\text{anch}}(a,r,u)$ for the lollipop specified by its anchor $a \in \mathbb{Q}^2$, radius $r \in \mathbb{Q}_{>0}$, and unit stem direction $u \in \mathbb{Q}^2$, and write $Q_{\text{cent}}(c,r,u)$ when the center is specified instead. Set

$$u_0 = (1,0), \qquad u_1 = \frac{1}{145}(144,-17), \qquad u_2 = \frac{1}{85}(-36,-77), \qquad u_3 = \frac{1}{145}(-17,144).$$

The identities $144^2 + 17^2 = 145^2$ and $36^2 + 77^2 = 85^2$ show that every u_i is a unit vector. Consider

$$\begin{aligned} Q_0 &= Q_{\text{anch}}\big((0,0), 200, u_0\big), \\ Q_1 &= Q_{\text{cent}}\big((9/20, 2/5), 11/20, u_1\big), \\ Q_2 &= Q_{\text{anch}}\big((23/20, 13/20), 100, u_2\big), \\ Q_3 &= Q_{\text{anch}}\big((21/20, -1/20), 100, u_3\big). \end{aligned}$$

This is a rational perturbation of Karlsson's four-lollipop construction.

Proposition 8.1 (exact rational base table)**.** *Every component intersection in the following table is transverse; the supporting stem lines are pairwise nonparallel; no listed intersection is an anchor; and the component counts are*

pair	$C_i \cap C_j$	$R_i \cap C_j$	$R_j \cap C_i$	$R_i \cap R_j$	*total*
Q_0, Q_1	2	2	0	1	5
Q_0, Q_2	2	2	2	1	7
Q_0, Q_3	2	2	2	1	7
Q_1, Q_2	2	2	2	1	7
Q_1, Q_3	2	2	2	1	7
Q_2, Q_3	2	2	2	1	7

Thus the base has forty crossings.

Proof. All coordinates are rational, so the verification reduces to strict rational inequalities. For an ordered pair i, j, put $d = c_j - c_i$, $p = d \cdot u_i$, and

$$\Delta_{ij} = r_j^2 - (|d|^2 - p^2), \qquad A_{ij} = |d - r_i u_i|^2 - r_j^2, \qquad T_{ij} = p - r_i.$$

The circle–circle component has two transverse points exactly when

$$(r_i - r_j)^2 < |d|^2 < (r_i + r_j)^2.$$

When $\Delta_{ij} > 0$, the supporting line meets the circle twice. The accepted ray component has one point if $A_{ij} < 0$, two points if $A_{ij} > 0$ and $T_{ij} > 0$, and no point if $A_{ij} > 0$ and $T_{ij} < 0$. These alternatives follow by evaluating the convex quadratic $|c_i + tu_i - c_j|^2 - r_j^2$ at $t = r_i$ and locating its vertex at $t = p$.

For the two supporting stem lines, put $D_{ij} = \det(u_i, u_j)$. When $D_{ij} \neq 0$, the parameters of their unique line intersection are

$$t_i = \frac{\det(d, u_j)}{D_{ij}}, \qquad t_j = \frac{\det(d, u_i)}{D_{ij}}.$$

It is a ray–ray point precisely when $t_i > r_i$ and $t_j > r_j$.

Substitution of the displayed rational data into these tests gives the six rows of the table. The ancillary program `scripts/certify_rational_base.py` performs exactly those substitutions using `fractions.Fraction`; its branches and sign decisions use no floating-point arithmetic. The full reduced fractions for every decisive expression are recorded in `verification/rational_base_verification.json`. In particular all circle inequalities are strict, every accepted mixed root is simple and away from its anchor, all D_{ij} are nonzero, and both ray–ray parameters are strictly beyond their anchor thresholds. The smallest absolute decisive quantity in this exact calculation is $151/23400 > 0$. $\square$

8.2 Pairwise chambers and a polynomial local family

Lemma 8.2 (pairwise chamber stability)**.** *Let L, M be a pair for which the circles are neither tangent nor coincident, every accepted circle–stem intersection is transverse and away from an anchor, the supporting stem lines are nonparallel, and their finite line intersection is not an anchor. Then there are parameter neighborhoods of L and M on which the number of intersections in each of the four component pairs is constant. All the stated strict conditions remain valid in those neighborhoods.*

Proof. Circle–circle alternatives are determined by strict inequalities involving the center distance and the sum and difference of radii. A mixed component is determined by a quadratic in the stem parameter. Simple accepted roots vary continuously. A double root can occur only outside the accepted half-line under the hypotheses; its positive distance from the anchor threshold persists under a small perturbation. Thus the number of accepted roots is locally constant. For the stems, the two supporting-line parameters are rational functions of the geometric parameters, with denominator the nonzero determinant of the direction vectors. Their strict positions relative to the two anchor thresholds persist. Intersecting the finitely many resulting open conditions proves the lemma. □

The following family replaces a previously proposed motion that produced only three intersections.

Lemma 8.3 (local four-crossing family)**.** *Let Λ be the unit lollipop centered at the origin with stem $\{(q,0) : q \geq 1\}$. There is a continuous family $(\Gamma_t)_{0\leq t\leq 1/4}$, with $\Gamma_0 = \Lambda$, such that every two distinct members have exactly four transverse crossings, none at an anchor.*

Proof. Set

$$c_t = (3t, 3t), \qquad v_t = (1-t^2, -2t), \qquad r_t = 1+t^2,$$

and let Γ_t have center c_t, radius r_t, and stem $\{c_t + qv_t : q \geq 1\}$. Since $|v_t|^2 = (1-t^2)^2 + 4t^2 = (1+t^2)^2 = r_t^2$, the stem is radial and outward.

Fix $0 \leq s < t \leq 1/4$ and put $\delta = t - s$. The squared center distance is $18\delta^2$, while the radius difference is $\delta(s+t)$. Hence

$$\delta^2(s+t)^2 < 18\delta^2 < (2+s^2+t^2)^2,$$

so the circles meet transversely twice.

For the stem of Γ_s against the circle of Γ_t, put

$$F_{s,t}(q) = |c_s + qv_s - c_t|^2 - r_t^2.$$

A direct expansion gives

$$F_{s,t}(1) = -\delta B_{s,t},$$

where

$$B_{s,t} = 6 + 8s - 16t - 6s^2 + s^3 + s^2t + st^2 + t^3 \geq 6 - 16t - 6s^2 \geq \frac{13}{8}.$$

Moreover

$$\frac{1}{2}F'_{s,t}(1) = (1+s^2)^2 - 3\delta(1-2s-s^2) \geq 1 - 3\delta \geq \frac{1}{4}.$$

Since $F''_{s,t} = 2|v_s|^2 > 0$, the function is strictly increasing on $[1,\infty)$, begins negative, and tends to infinity. It has exactly one simple zero there.

In the reverse direction, define

$$G_{s,t}(q) = |c_t + qv_t - c_s|^2 - r_s^2.$$

Then

$$G_{s,t}(1) = \delta C_{s,t}, \quad C_{s,t} = 6 - 16s + 8t - 6t^2 + s^3 + s^2 t + st^2 + t^3 > 6 - 8t - 6t^2 \geq \frac{29}{8},$$

and

$$\frac{1}{2} G'_{s,t}(1) = (1+t^2)^2 + 3\delta(1 - 2t - t^2) > 0.$$

Thus $G_{s,t}$ is positive and increasing on $[1, \infty)$, so this mixed component is empty.

Finally,

$$\det(v_s, v_t) = 2(s-t)(1+st) \neq 0.$$

Solving $c_s + \lambda v_s = c_t + \mu v_t$ gives

$$\lambda = \frac{3(1+2t-t^2)}{2(1+st)}, \qquad \mu = \frac{3(1+2s-s^2)}{2(1+st)}.$$

Both exceed one, because

$$1 + 6t - 3t^2 - 2st > 0, \qquad 1 + 6s - 3s^2 - 2st > 0;$$

the negative terms in either expression have total magnitude at most 5/16. Hence the stems meet once beyond both anchors. A point strictly beyond an anchor lies outside its own circle, which also prevents coincidence of this point with any circle crossing and prevents the unique mixed point from being a triple component intersection. The total is $2 + 1 + 0 + 1 = 4$. □

Lemma 8.4 (genericization inside a chamber)**.** *Let a finite lollipop arrangement have transverse pairwise crossings away from anchors and pairwise nonparallel supporting stem lines. Every sufficiently small product neighborhood preserving its pairwise chambers contains an arrangement with no finite triple point and with all pairwise component counts unchanged.*

Proof. Shrink to a product of connected coordinate balls on which Lemma 8.2 preserves every pairwise count and strict condition. Choose the lollipops successively. After $k-1$ have been chosen generically, their finite pairwise intersection set P is finite. For a fixed $p \in P$, the parameters (c, r, θ) for which the next circle contains p lie in the zero set of the nonzero real-analytic function

$$g_p(c, r, \theta) = |p - c|^2 - r^2.$$

The parameters for which its supporting stem line contains p lie in the zero set of the nonzero real-analytic function

$$h_p(c, r, \theta) = \det(p - c, u_\theta).$$

Neither function vanishes identically on a coordinate ball: vary r for g_p, and vary c transversely to u_θ for h_p. A proper real-analytic zero set is closed with empty interior, hence nowhere dense. A finite union of these closed nowhere-dense sets cannot contain the open parameter ball. Choose the k-th lollipop outside that union. It avoids all old pairwise intersection points, while the chamber conditions retain all pairwise counts, transversality, nonparallelity, and anchor exclusions. Induction removes every finite triple point. □

Lemma 8.5 (four-cluster blow-up)**.** *For any nonnegative integers m_0, m_1, m_2, m_3, there is a generic arrangement of $n = \sum_i m_i$ lollipops having*

$$4 \sum_i \binom{m_i}{2} + 5 m_0 m_1 + 7 \sum_{\substack{i<j \\ (i,j) \neq (0,1)}} m_i m_j \tag{23}$$

finite crossings.

Proof. By Proposition 8.1 and Lemma 8.2, choose neighborhoods $\mathcal{U}_i$ of Q_i such that every cross-cluster pair in $\mathcal{U}_i \times \mathcal{U}_j$ has the base component counts and nonparallel stems. Map the local family of Lemma 8.3 to Q_i by a similarity. A sufficiently short initial interval lies in $\mathcal{U}_i$. Choose m_i distinct parameters from it. Same-cluster pairs have four crossings, while cross-cluster pairs have five for $(0, 1)$ and seven otherwise. All pairwise intersections are transverse and away from anchors. Apply Lemma 8.4 in a sufficiently small product chamber to remove accidental triple points without changing any pair count. Summing gives (23). □

For $n = \sum_i m_i$, subtracting $4\binom{n}{2}$ from (23) gives

$$3 \sum_{i<j} m_i m_j - 2m_0 m_1.$$

For a fixed multiset of cluster sizes, this is maximized by assigning the two smallest sizes to m_0, m_1. Maximizing over all quadruples gives $S(n)$. The final arrangement is generic, so the equality case of Proposition 2.1 gives

$$a_{\mathcal{L}}(n) \geq 4\binom{n}{2} + S(n) + n + 1.$$

Together with the upper bound already proved, this completes the proof of Theorem 1.1.

9 Asymptotics and values

The continuous relaxation of (15) has its unique minimum at

$$a = b = \frac{3n}{16}, \qquad c = d = \frac{5n}{16}.$$

Rounding changes the quadratic objective by $O(1)$, so

$$M(n) = \frac{15}{32}n^2 + O(1), \qquad S(n) = \frac{33}{32}n^2 + O(1),$$

and therefore

$$a_{\mathcal{L}}(n) = \frac{97}{32}n^2 - n + O(1).$$

The first values are shown below.

n	one optimal (a,b,c,d)	$S(n)$	$a_{\mathcal{L}}(n)$
0	(0,0,0,0)	0	1
1	(0,0,0,1)	0	2
2	(0,0,1,1)	3	10
3	(0,1,1,1)	9	25
4	(1,1,1,1)	16	45
5	(1,1,1,2)	25	71
6	(1,1,2,2)	37	104
7	(1,2,2,2)	50	142
8	(1,2,2,3)	65	186
9	(1,2,3,3)	83	237
10	(2,2,3,3)	103	294
11	(2,2,3,4)	124	356
12	(2,2,4,4)	148	425
13	(2,3,4,4)	174	500
14	(2,3,4,5)	201	580
15	(2,3,5,5)	231	667
16	(3,3,5,5)	264	761
17	(3,3,5,6)	297	859
18	(3,3,6,6)	333	964
19	(3,4,6,6)	372	1076
20	(4,4,6,6)	412	1193

Reproducibility statement

The ancillary bundle contains the complete LaTeX source, the exact rational verifier for Proposition 8.1, an independent finite checker for the colored and matrix extremal problems, their machine outputs, and SHA-256 checksums. The base verifier uses no floating-point branch or sign decision. The finite extremal checks are corroborative only; the proofs in Sections 5–7 are symbolic and do not depend on them.

References

[1] D. O. H. Cutler, J. Karlsson, and N. J. A. Sloane, *Cutting a Pancake with an Exotic Knife*, arXiv:2511.15864v3, 2026.

[2] S. Mahajan and P. Chopra, *A Two-Graph Refinement of Paulsen's Lollipop Bounds*, arXiv:2606.06064, 2026.

[3] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A389624, 2026.

[4] M. Paulsen, *The Lollipop Problem*, 2026, https://math.mp42.de/lollipop/.

[5] J. Bochnak, M. Coste, and M.-F. Roy, *Real Algebraic Geometry*, Springer, 1998.

[6] A. Hatcher, *Algebraic Topology*, Cambridge University Press, 2002.

[7] P. Turán, On an extremal problem in graph theory, *Matematikai es Fizikai Lapok* **48** (1941), 436–452.

[8] A. A. Zykov, On some properties of linear complexes, *Matematicheskii Sbornik* **24** (1949), 163–188.